

Group: HOOMANS

Phase A: Foundation & Question Development

Research Question:

To what extent use of software-based “Blue Light Filtering” and Display mode affect user eye strain and comfort?

Background/ Development:

Many users think that using their devices in dark mode helps protecting their eyesight, while some rely on software that claim to reduce blue light radiations from the screen. Individuals who spend more time on screens are more concerned about these factors. So, this study investigates whether dark mode and blue light filter are effective in reducing eye strain; do they actually work or not?

Relationships we are interested in:

- Display mode to Eye Strain
- Blue- Light filter to Eye Strain
- Screen Exposure Duration to Eye Strain
- Use of Special Glasses to Eye Strain

Common Variables:

- Independent:
 - Display Mode
 - Blue Light Filter Status
 - Use of Special Glasses
- Dependent:
 - Eye Strain (will be measured on scale from 1-10)
- Control Variables:
 - Screen Brightness Level
 - Duration of exposure
 - Device type
 - Posture / viewing distance
 - Environmental Variables
 - Type of activity on device
- User Related Variables:
 - Age
 - Vision Condition
 - Daily Average Screen Time
 - History of Visual Fatigue